

AAGB - TME 5: Networks

1 Introduction

The theory of networks involves the study of statistical properties that characterize the structure and behavior of systems defined by networks. We aim to study some classical measures for networks: degree distribution, clustering coefficient, and betweenness centrality. To test your functions, you can use the two files (**network1**, **network2**) which list all the edges present in a network. For example, these files can be parsed to create a dictionary where the keys are the nodes of the network and the values are the lists of nodes connected to each key node.

2 Biological Context

A graph is a mathematical structure that models relationships between pairs of objects. This structure is widely used in biology, allowing for the study of:

- Protein-protein interaction networks (see figure, where each node represents a protein, and each edge represents an interaction between two proteins).
- Metabolic networks (the set of metabolic and physical processes that determine the physiological and biochemical properties of a cell).
- Genetic expression networks (where each node corresponds to a gene and each edge to a co-expression relationship).
- Phylogenetic networks (used to study evolutionary relationships).

Studying these graphs using measures like those described below is important because it allows us to analyze network properties, for example, by predicting the function of unknown elements based on their relationships with well-characterized elements.

Most of these networks are not random but *scale-free*, as in other contexts such as social networks, the internet, etc. These networks have the following characteristics:

- Robustness (a good ability to withstand random attacks).
- Heterogeneity (hubs, i.e., nodes with the most edges, are generally not connected to other hubs).
- High internal connectivity within the graph, with few connections to the periphery.

3 Degree Distribution

1. Define the degree distribution for a graph.
2. What is the significance of studying the degree distribution in a graph?
3. Plot the degree distribution of the given networks.
4. What is a scale-free network? Among the two networks studied, which could be scale-free and which could be random? Why?

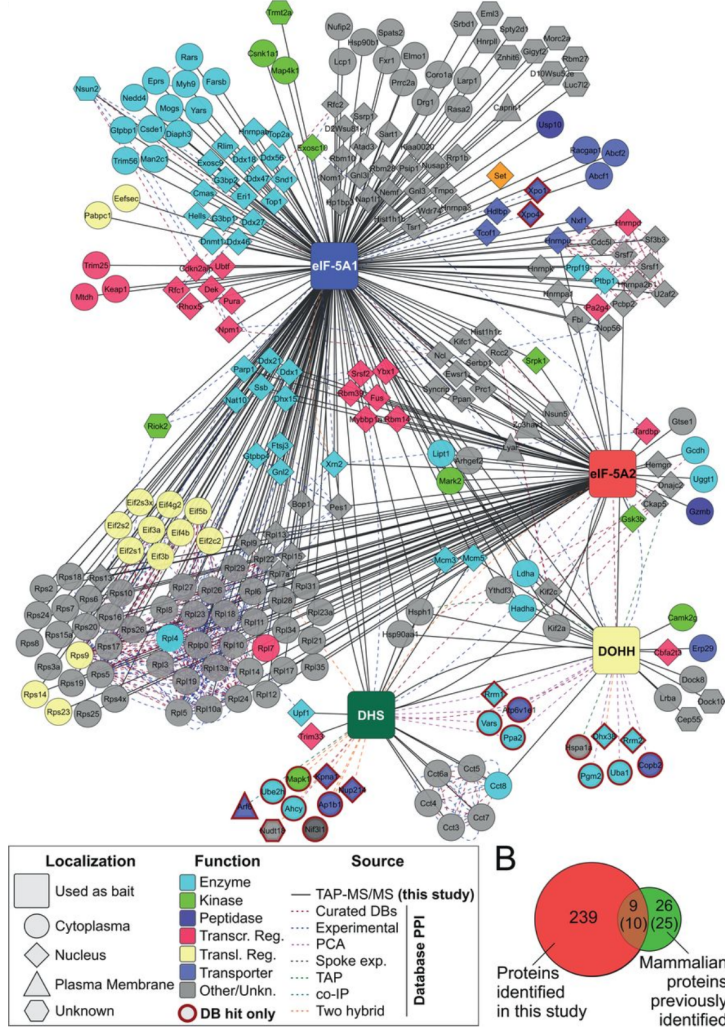


Figure 1: Protein-protein-interaction Network Organization of the Hypusine Modification System (H. Sievert et al.).

4 Clustering Coefficient

The clustering coefficient provides a local measure of the interconnectivity of a node's neighbors. For a scale-free network, the log-log distribution of average clustering coefficients, $C(k)$, as a function of node degree k , follows a power law of the form $C(k) \propto k^{-\gamma}$. This indicates that some network nodes form clusters (in contrast to random networks).

Definition 1 (Clustering Coefficient)

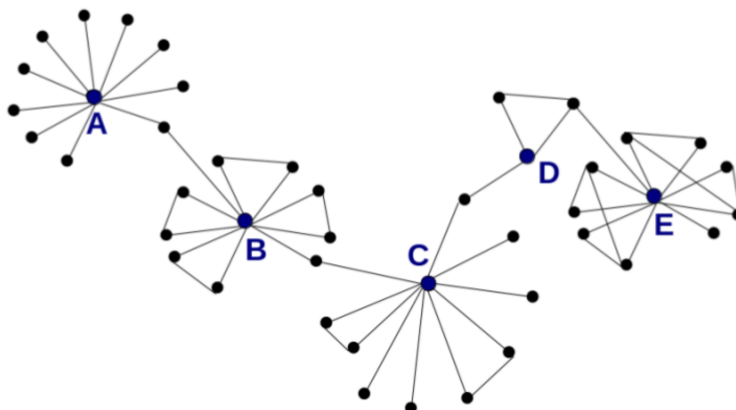
Let $G = (V, E)$ be an undirected graph and $v \in V$ a node. Let $N(v)$ be the set of neighbors of v , and $E(N(v)) = \{(u_1, u_2) \in E \mid u_1, u_2 \in N(v)\}$.

The clustering coefficient $cc(v)$ of v is defined as:

$$cc(v) = \frac{2|E(N(v))|}{d^o(v)(d^o(v) - 1)}$$

In other words, $cc(v)$ is the ratio between the number of "triangles" involving node v ($|E(N(v))|$) and the number of "triangles" that could potentially involve v ($d^o(v)(d^o(v) - 1)/2$).

Consider the following network G :



1. Calculate $cc(A)$, $cc(B)$, $cc(C)$, $cc(D)$, and $cc(E)$.
2. Write a function to calculate the average clustering coefficient. What can you conclude about the studied networks?

5 Betweenness Centrality

Betweenness centrality measures the importance of a node (or an edge) in facilitating the propagation of a signal/message, i.e., its role as an intermediary in the network structure. It relies on the notion of the shortest path (the path with the fewest edges).

Conventionally, let σ_{st} denote the number of shortest paths between nodes s and t in a graph G , and $\sigma_{st}(v)$ the number of shortest paths between s and t that pass through v (assuming $s \neq t \neq v$). The betweenness centrality of a node v is then defined as:

$$BC(v) = \sum_{s \neq t \neq v} \frac{\sigma_{st}(v)}{\sigma_{st}}$$

A simple case for calculating $BC(v)$ arises when there is exactly one shortest path for every pair of vertices (s, t) in G . In this case, $BC(v)$ is simply:

$$BC(v) = \sum_{s \neq t \neq v} \sigma_{st}(v),$$

i.e., the number of shortest paths in G that pass through v .

1. What type of nodes does betweenness centrality highlight? Is a network with a few nodes of high centrality robust? How can this measure be used to detect communities?
2. Write a function to calculate betweenness centrality.